



Flood Early Warning System for Chennai & Lessons from 2015 floods for Making Chennai Water Resilient



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Outline

✚ Chennai Floods 2015

✚ Rainfall

- Historic Intensity – Duration – Frequency curve
- A perspective of Dec.1, 2015 rainfall

✚ Landuse – Rapid Urbanization

✚ Reservoir Release

✚ Drainage System

✚ Tidal Levels

✚ Way Forward

✚ Salient Features of a Flood Resilient City

- Forecasting and Planning

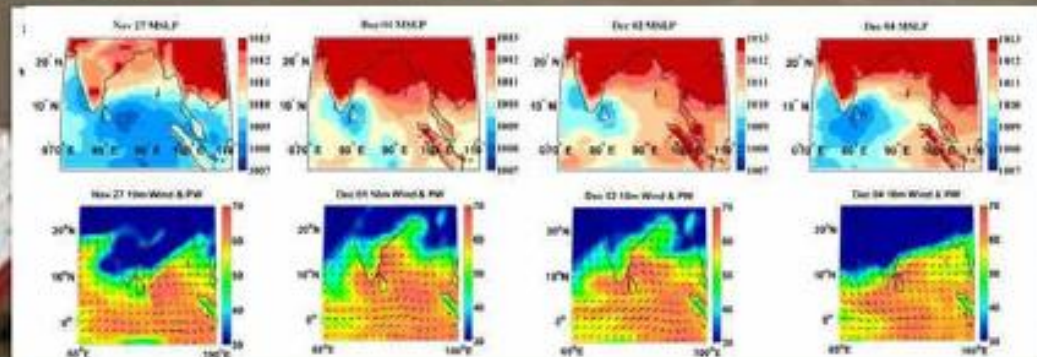
✚ Tangible actions



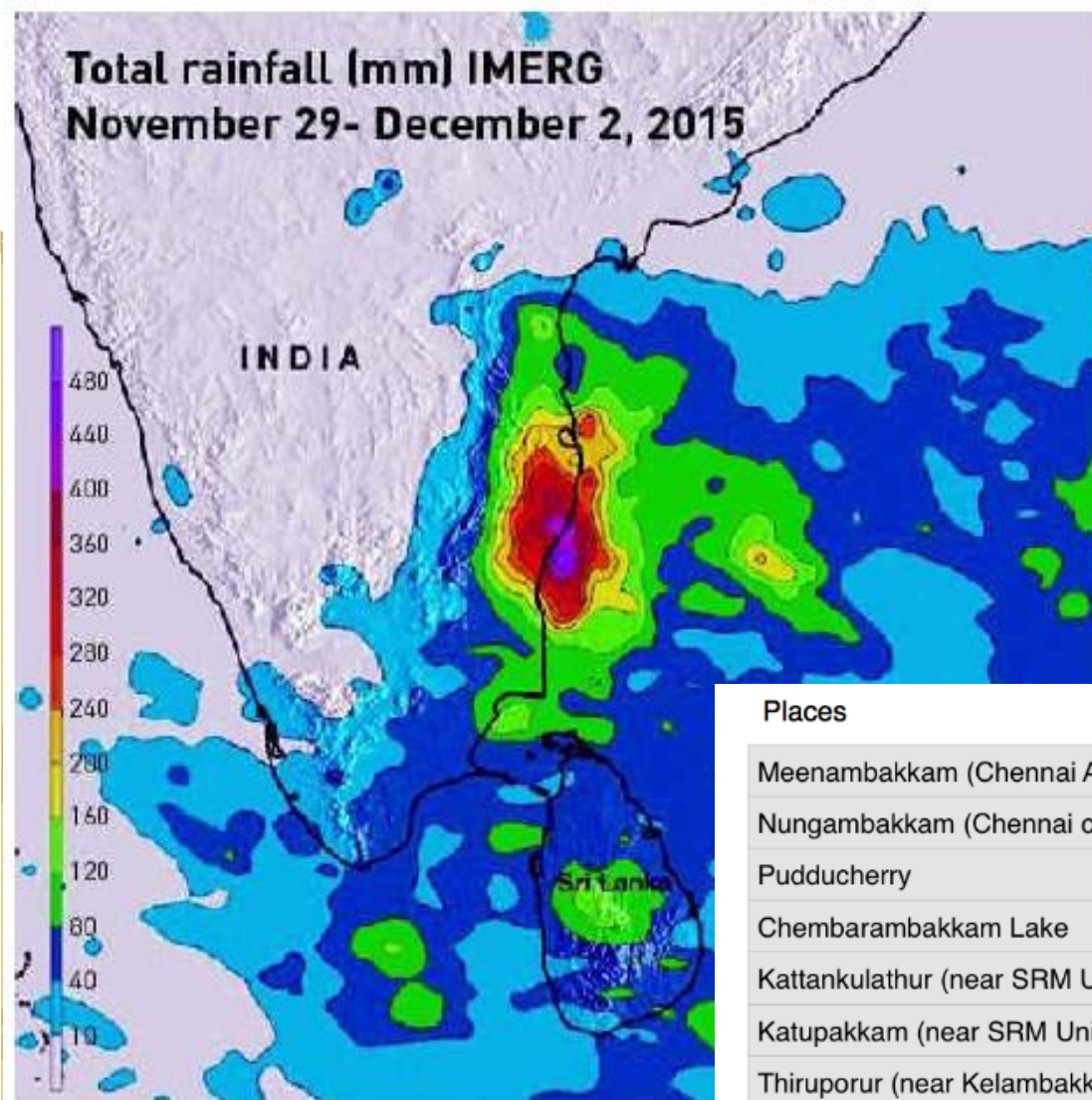
Chennai Floods 2015

A Rapid Assessment

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S. Murty Bhallamudi, IIT Madras
Arpita Mondal, IIT Bombay
Subimal Ghosh, IIT Bombay
Pradeep Mujumdar, IISc Bangalore



Total rainfall (mm) IMERG November 29- December 2, 2015



Dec.1-Dec.2, 2015: 24hr rainfall

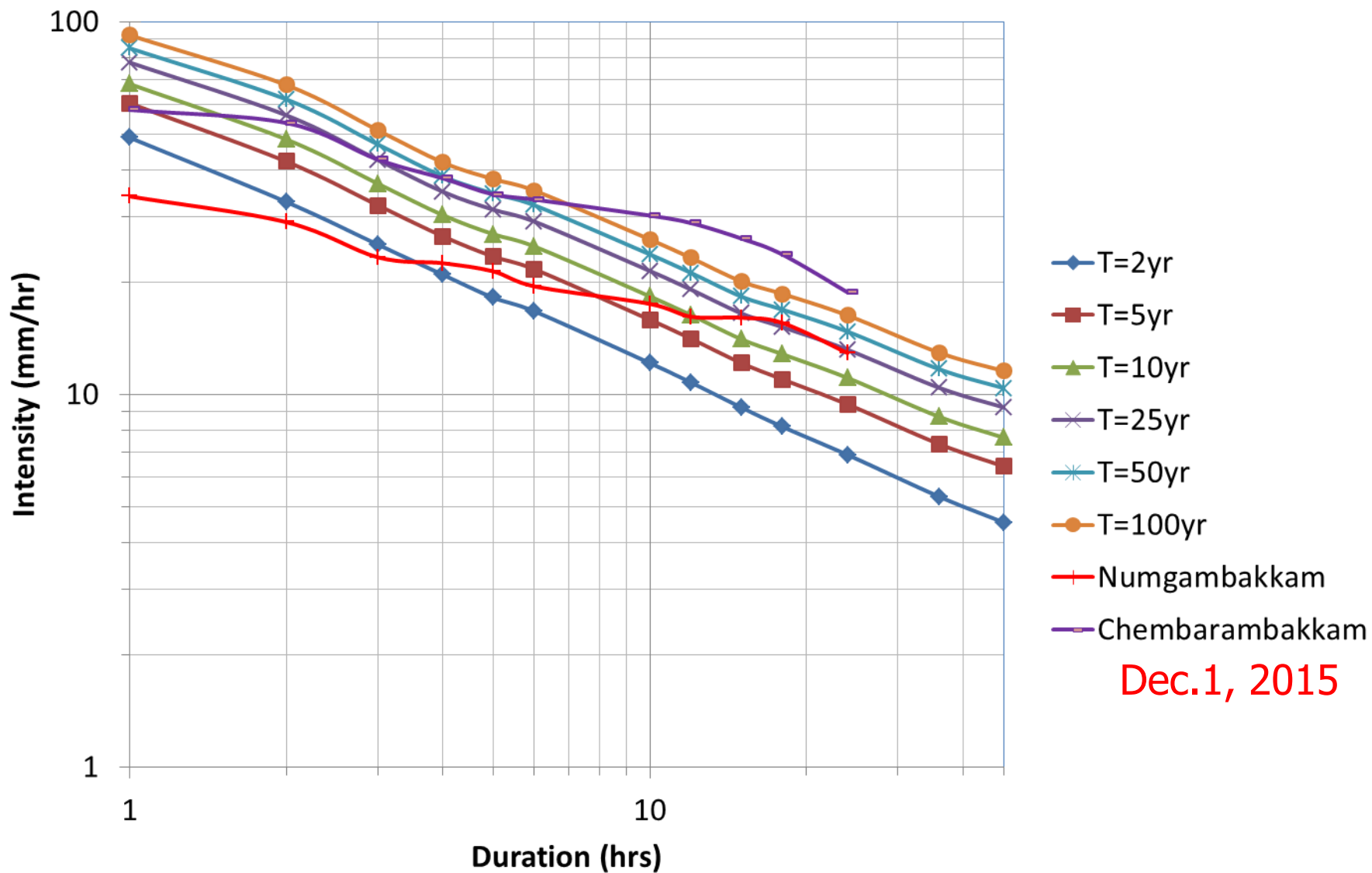
Places

Rainfall in millimetre

Meenambakkam (Chennai Airport)	345
Nungambakkam (Chennai city)	294
Pudducherry	218
Chembarambakkam Lake	475
Kattankulathur (near SRM University)	445
Katupakkam (near SRM University)	429
Thiruporur (near Kelambakkam)	348
Korattur Anaicut (Poonamalle)	336
Puzhal Agro	325
Redhills Lake	320
Anna University	319
Kolapakkam (near Chennai Airport)	310
Taramani	300

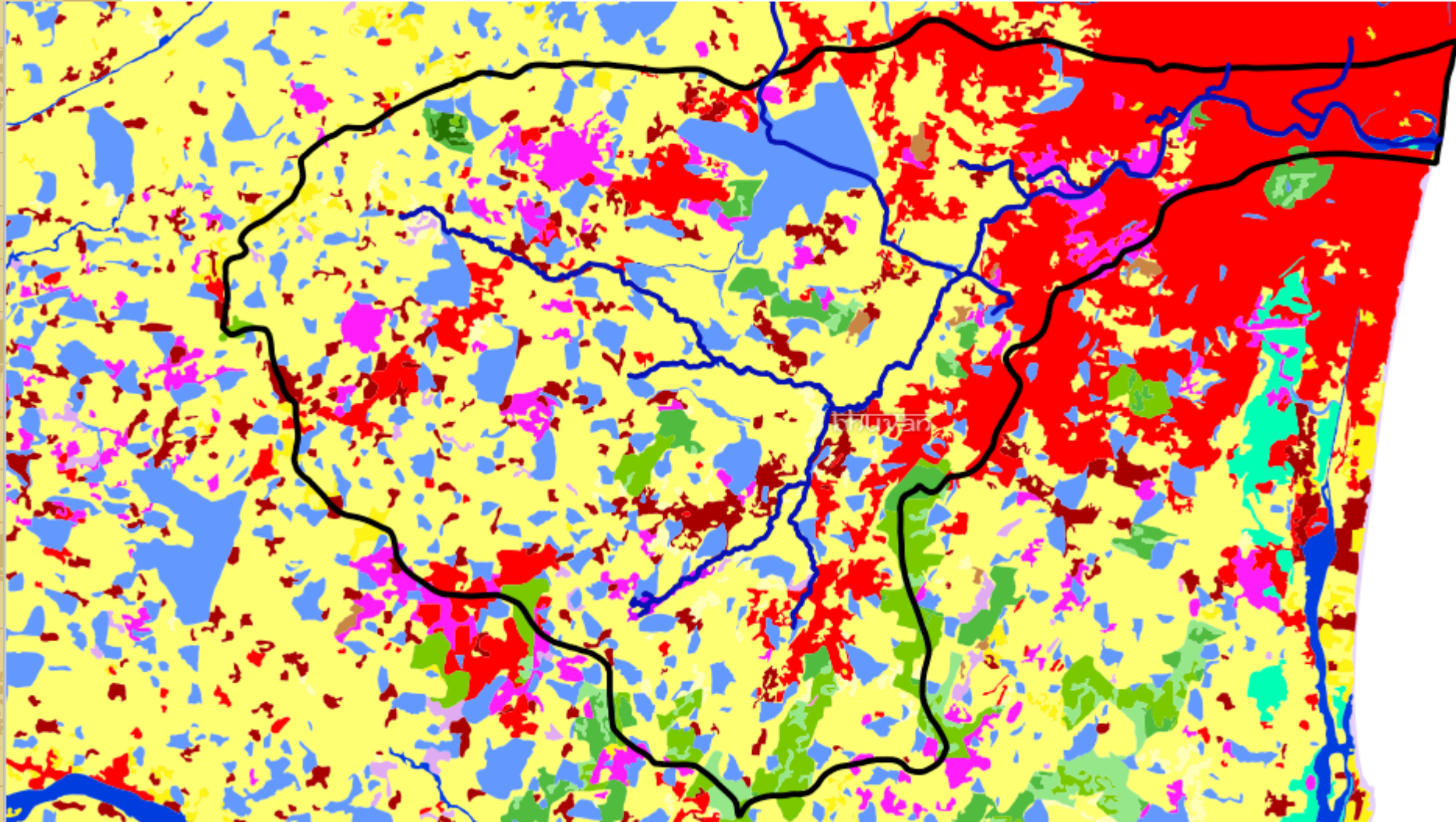
Chennai

Rainfall Intensity-Duration-Frequency





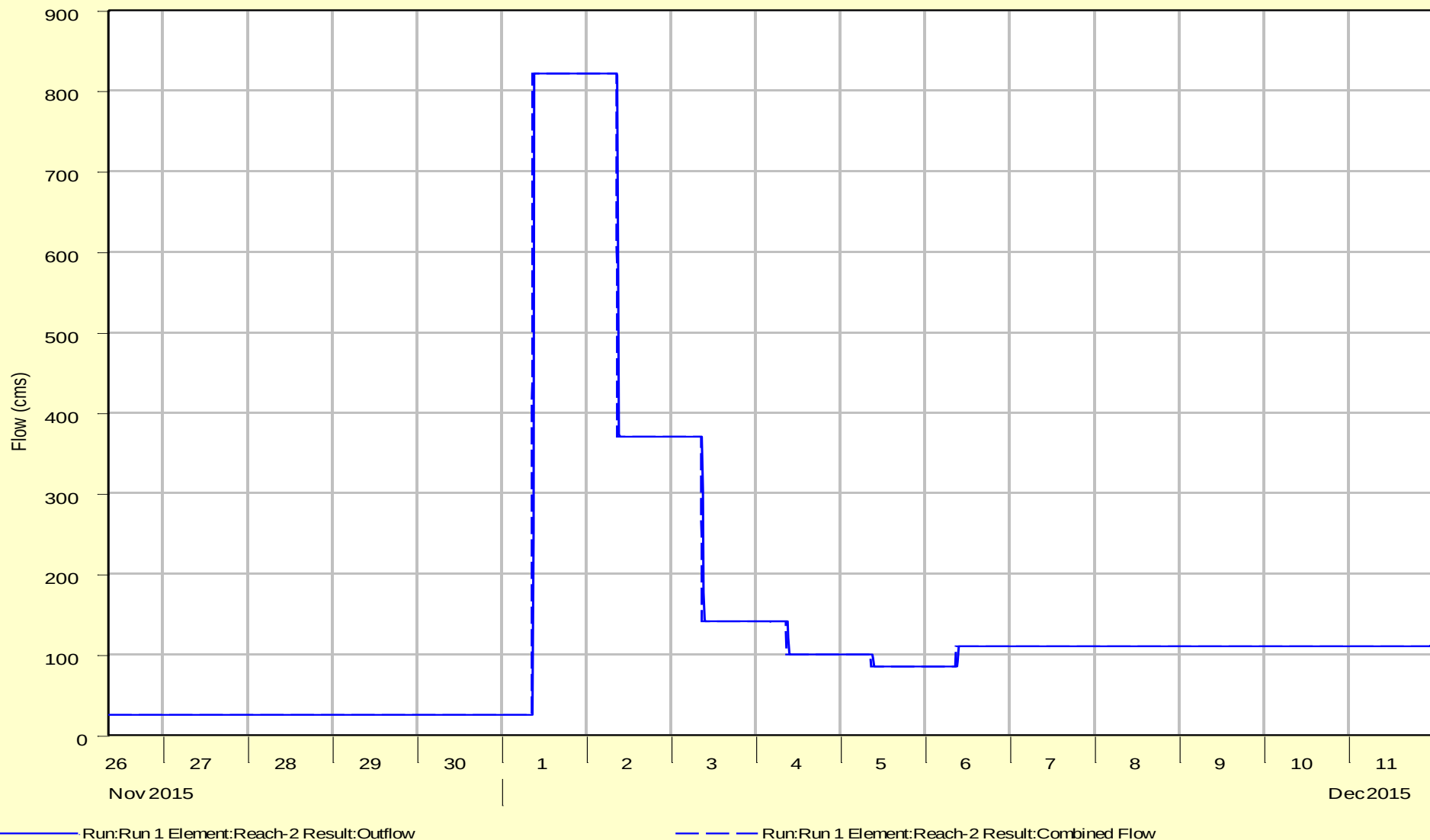
Landuse





Chembarambakkam Release

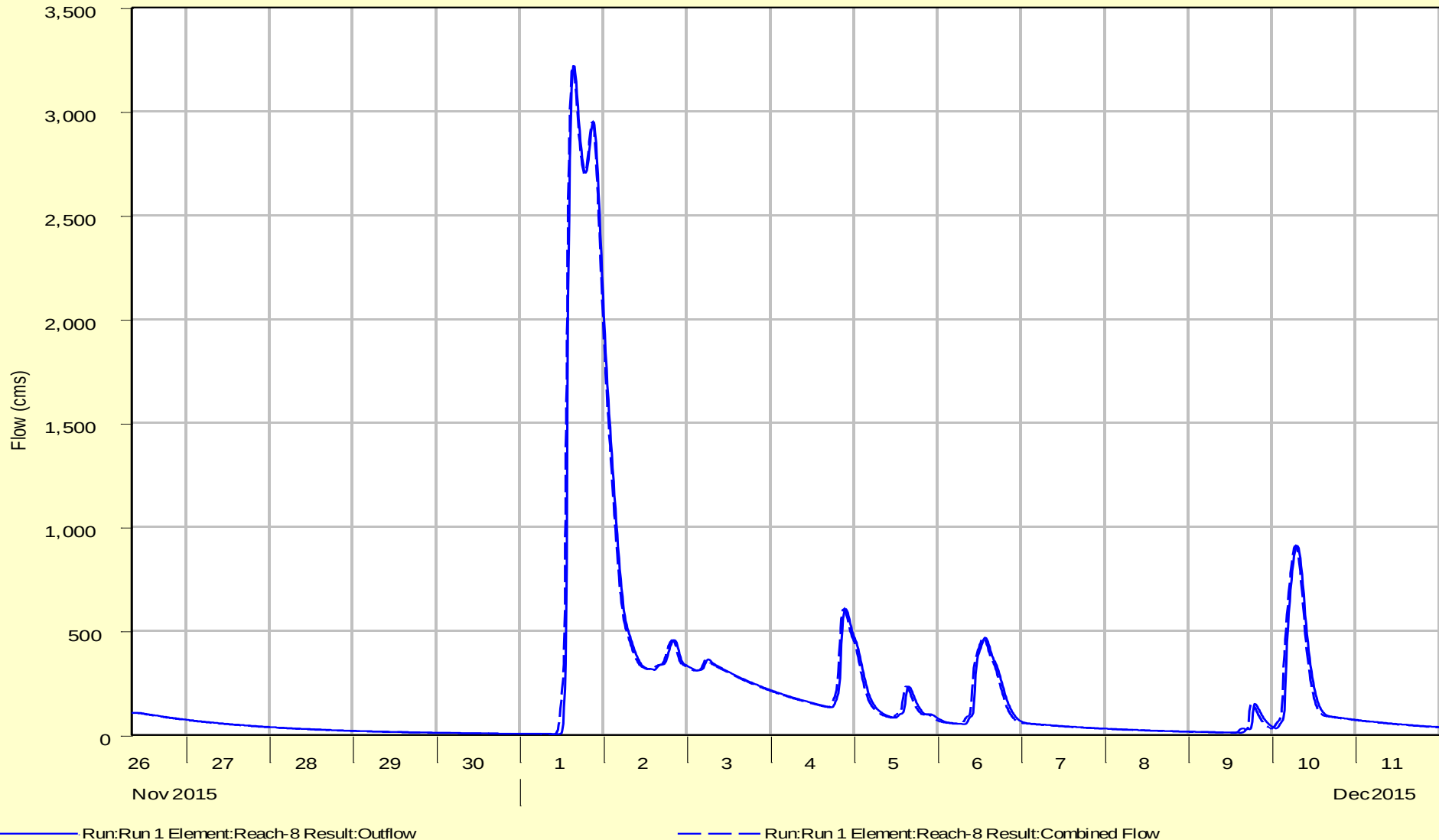
Reach "Reach-2" Results for Run "Run 1"





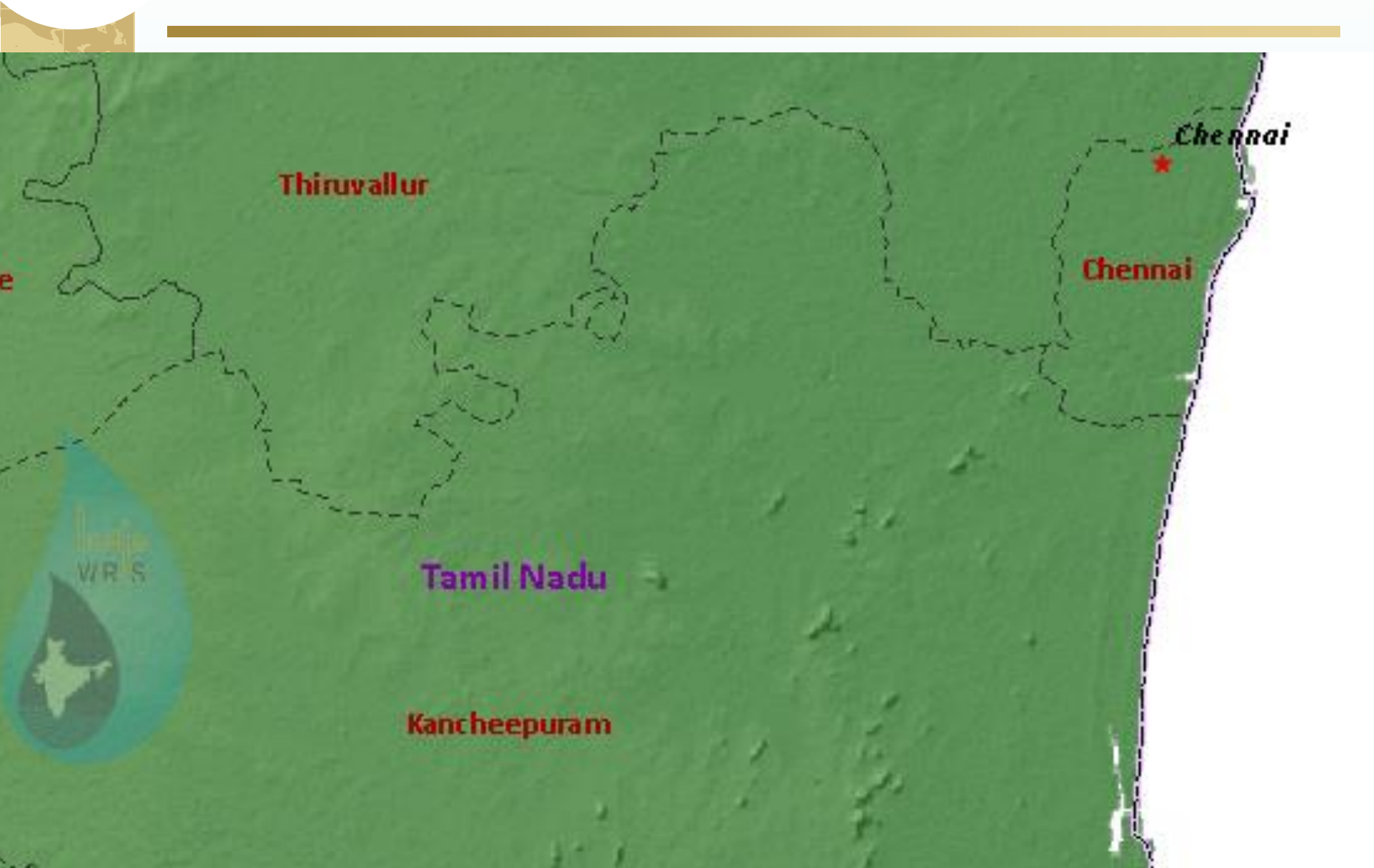
Estimated unregulated flow from the rest of the watershed

Reach "Reach-8" Results for Run "Run 1"



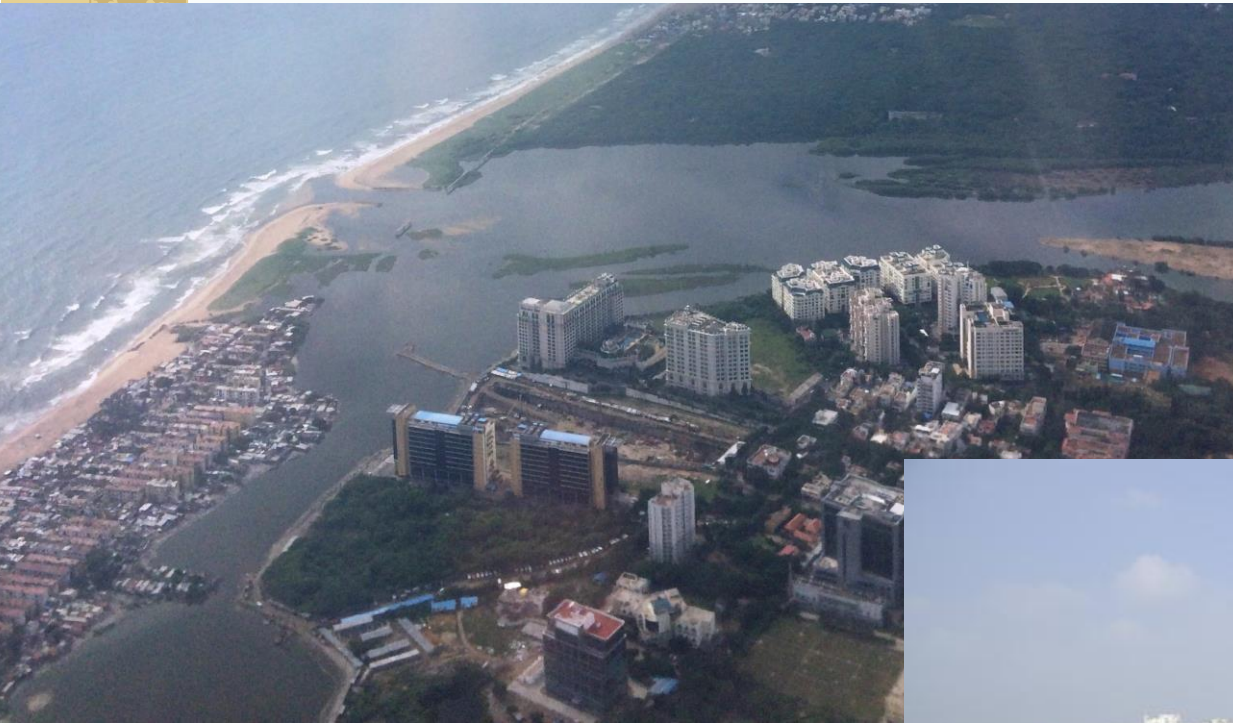


Topography: Less relief





Littoral Sediment and closure of river mouths



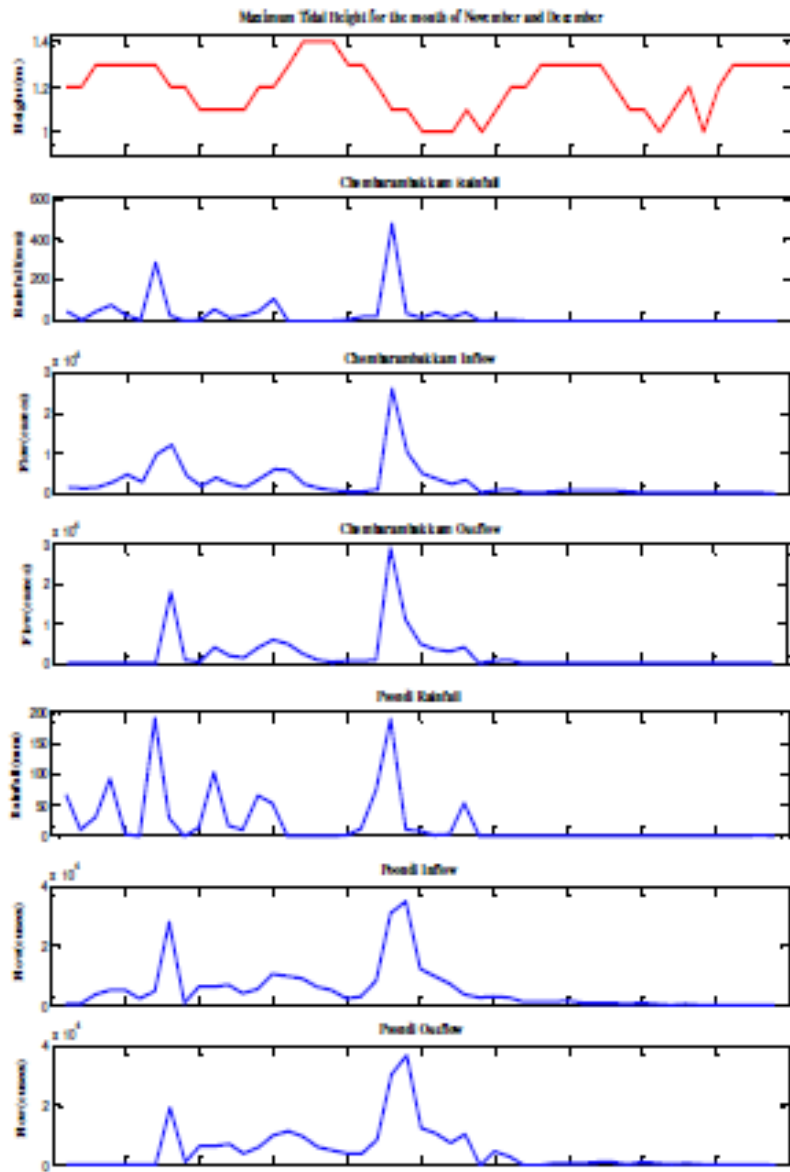


Figure 20: Maximum Tidal levels, Rainfall and Flows during November-December 2015
(Source: <http://www.tides4fishing.com/as/india/chennai> and Chennai Metropolitan Water Supply & Sewerage Board). 1 cumec = 0.028 cubic meters/sec.

Tidal levels and Reservoir Releases

Flood peak occurred during the low tide, otherwise the inundation extent could have been severe



Stormwater Drainage System

- ✚ Road network length: 6000 km
 - ✚ only about 1660 km of storm drains
 - Prior to 2015
 - ✚ 205 km of drains have a width of 0.6 m or more
 - ✚ Storm Water Drains are provided only for roads with 12 m width or more





Stormwater Drainage System

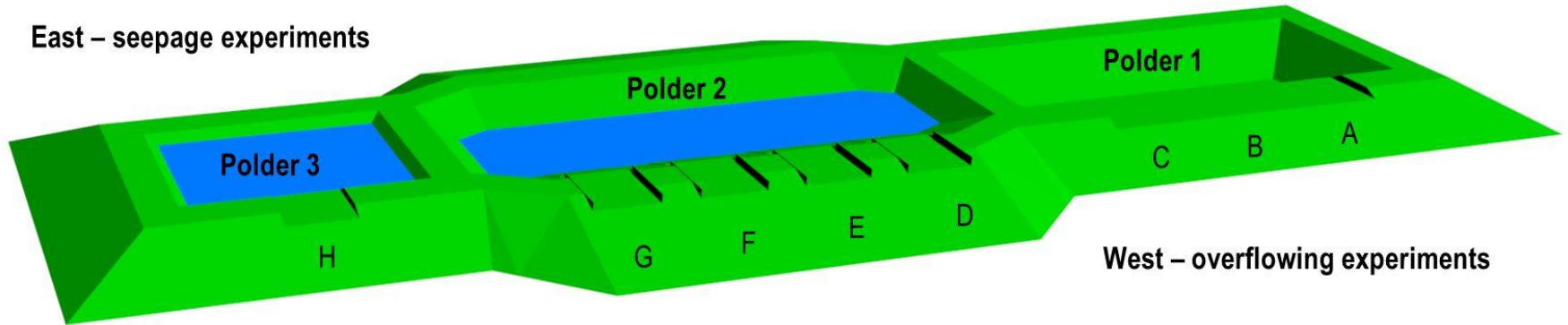
- ✚ Insufficient coverage with storm water drains
- ✚ Lack of Proper connectivity
 - ✚ Linkage to major canals and waterways
- ✚ Insufficient capacity
 - ✚ Original design intensity of 31.39 mm/h
 - 1hr storm duration and 2yr return period
 - Revised to 68 mm/h in 2014 (new drains)





Polder Effect of Road Aggradation

East – seepage experiments



West – overflowing experiments



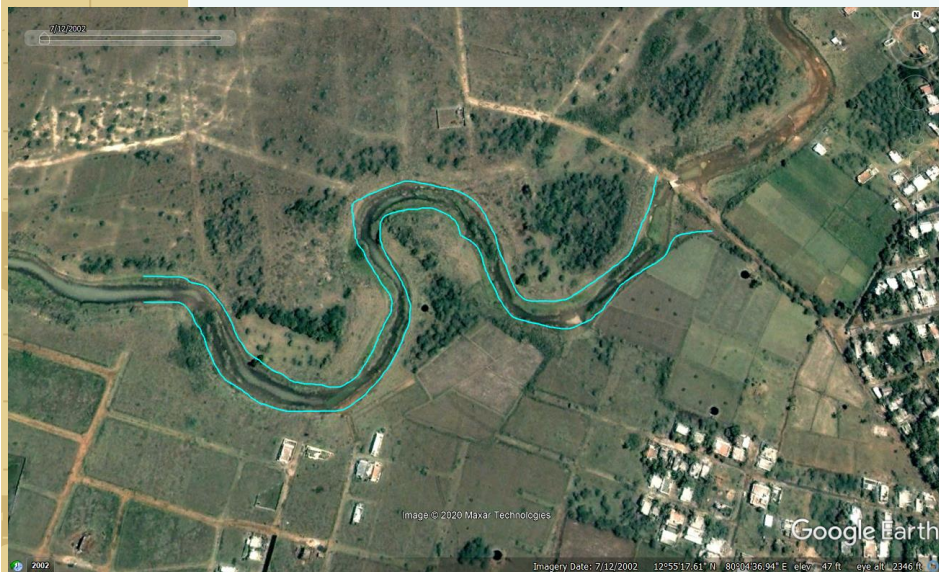
⊕ Roads fragment the natural drainage pathways

⊞ **Lack of adequate cross drainage works along roads**



Outer Ring Road (ORR) near Varadharajapuram

2002



Before the construction of Outer Ring Road

2020



After the construction of Outer Ring Road



How did Chennai 2015 flood happen?

- ⊕ Is climate and extreme weather the only cause?
 - ⊗ Only Partly;
 - ⊗ Topography, littoral sediment
 - ⊗ Design issues
- ⊕ Could this have been prevented?
 - ⊗ No
- ⊕ Could the effect have been minimized with a resilient design and management?
 - ⊗ Yes
- ⊕ Could this recur in the future?
 - ⊗ YES. Very likely. Could be even worse if we remain unprepared





Salient Features of a Flood Resilient City

Forecasting and Planning





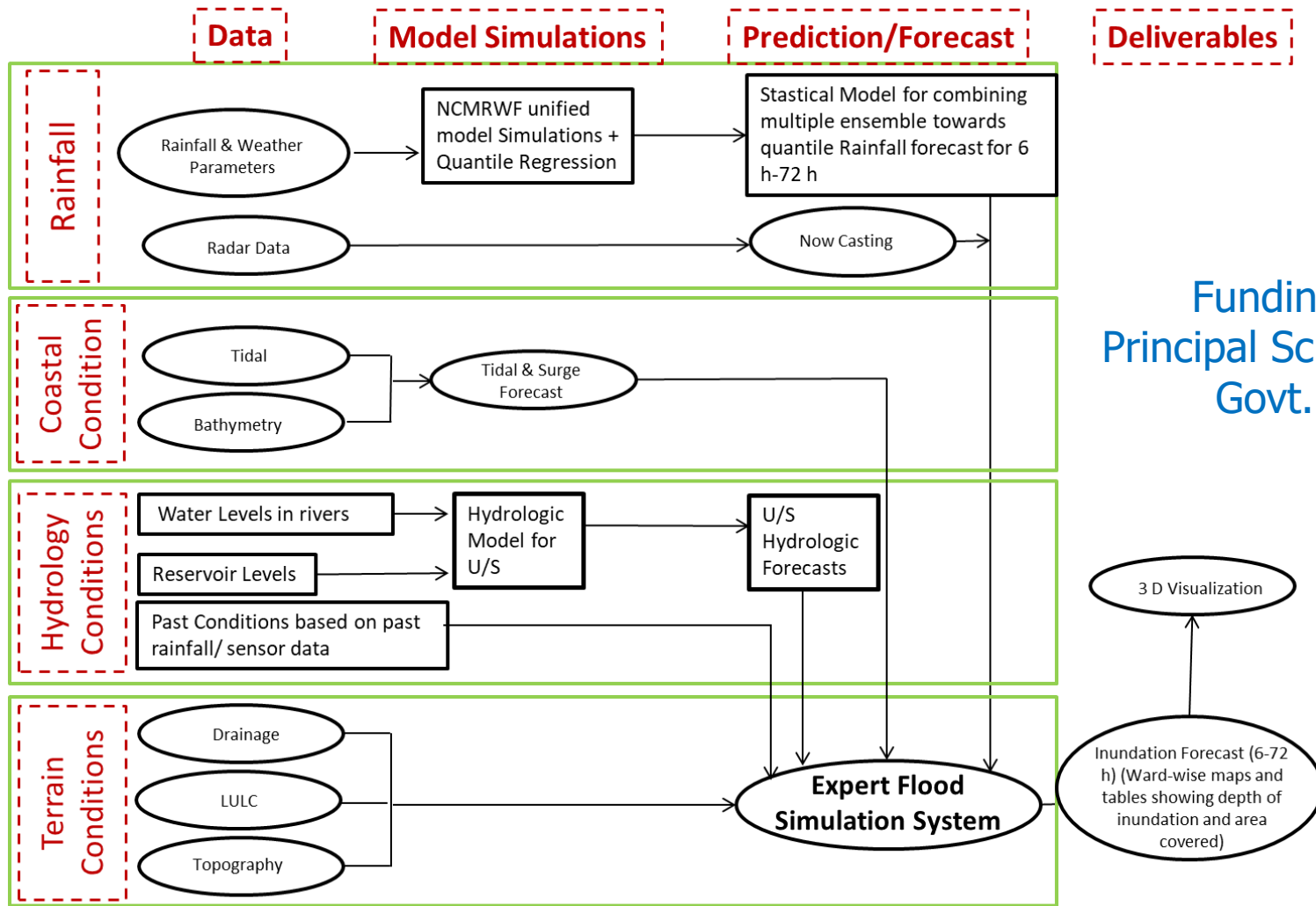
Resilience

- ✚ Capability to prepare for, respond to, and recover from significant multi-hazard threats with minimum damage to public safety and health, the economy, and security



Design of an Expert System for Flood Forecasting and Management for the city of Chennai

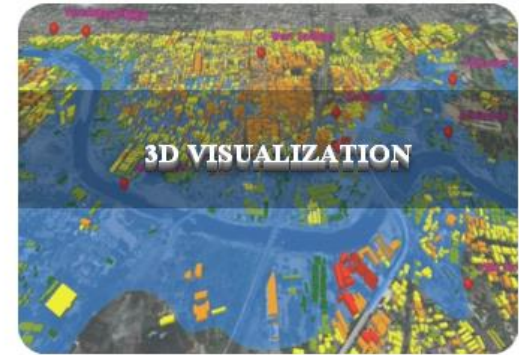
Framework



Funding support
Principal Scientific Advisor
Govt. of India

Fig. 1: Framework of the Flood Forecasting System of Chennai

Chennai Flood Warning System Dashboard

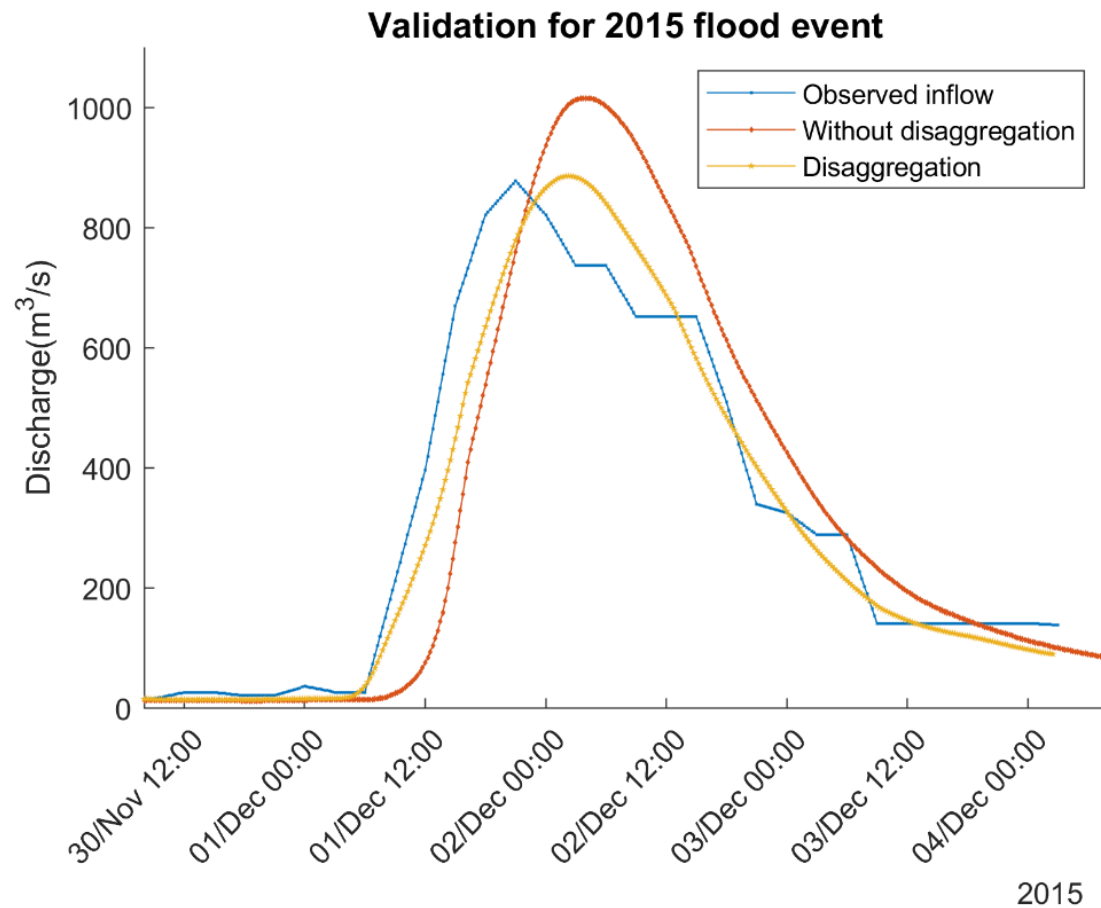


- A disaster preparedness decision support system for coastal flooding in urban areas

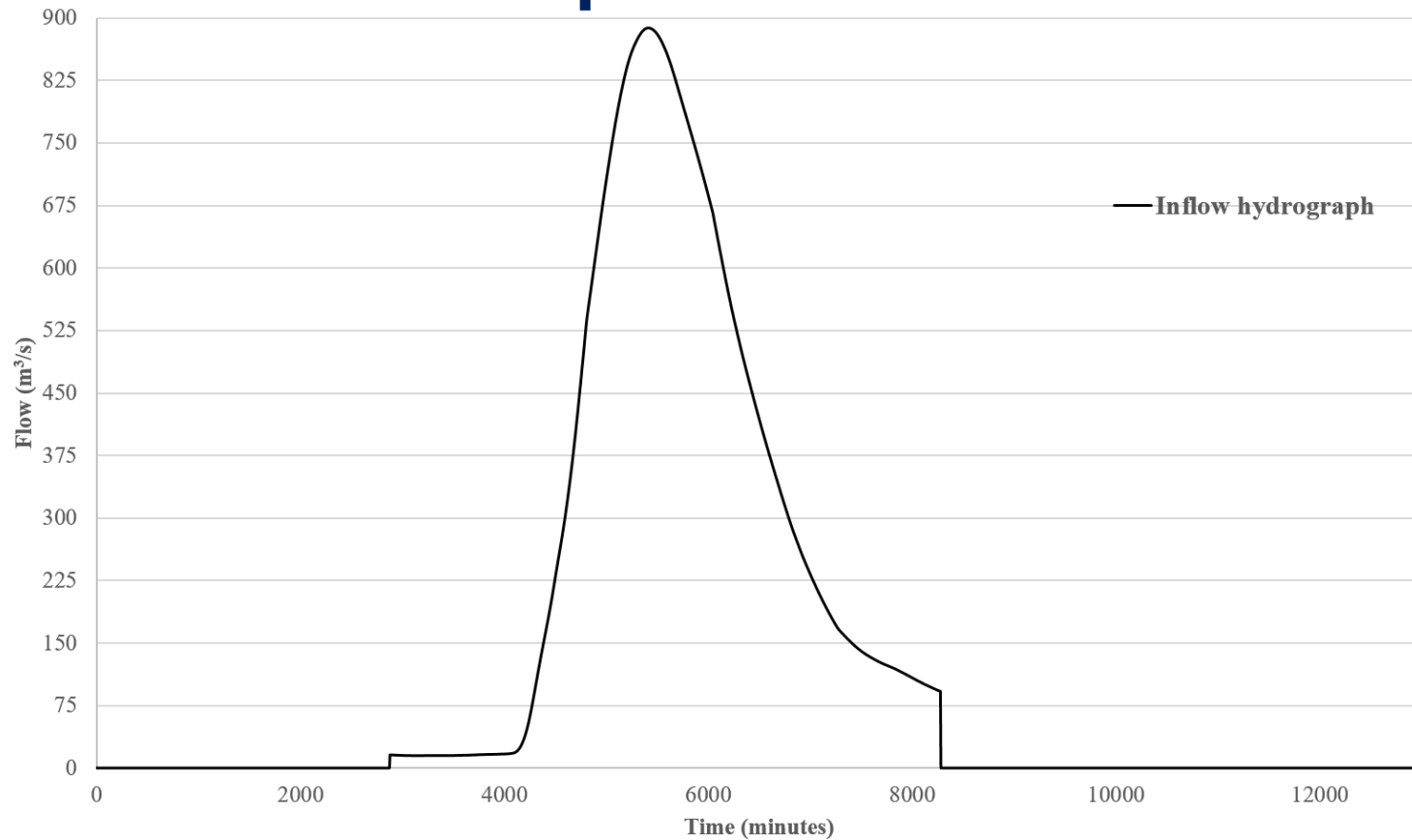
Water level logger, Automatic water level recorder, ADVP



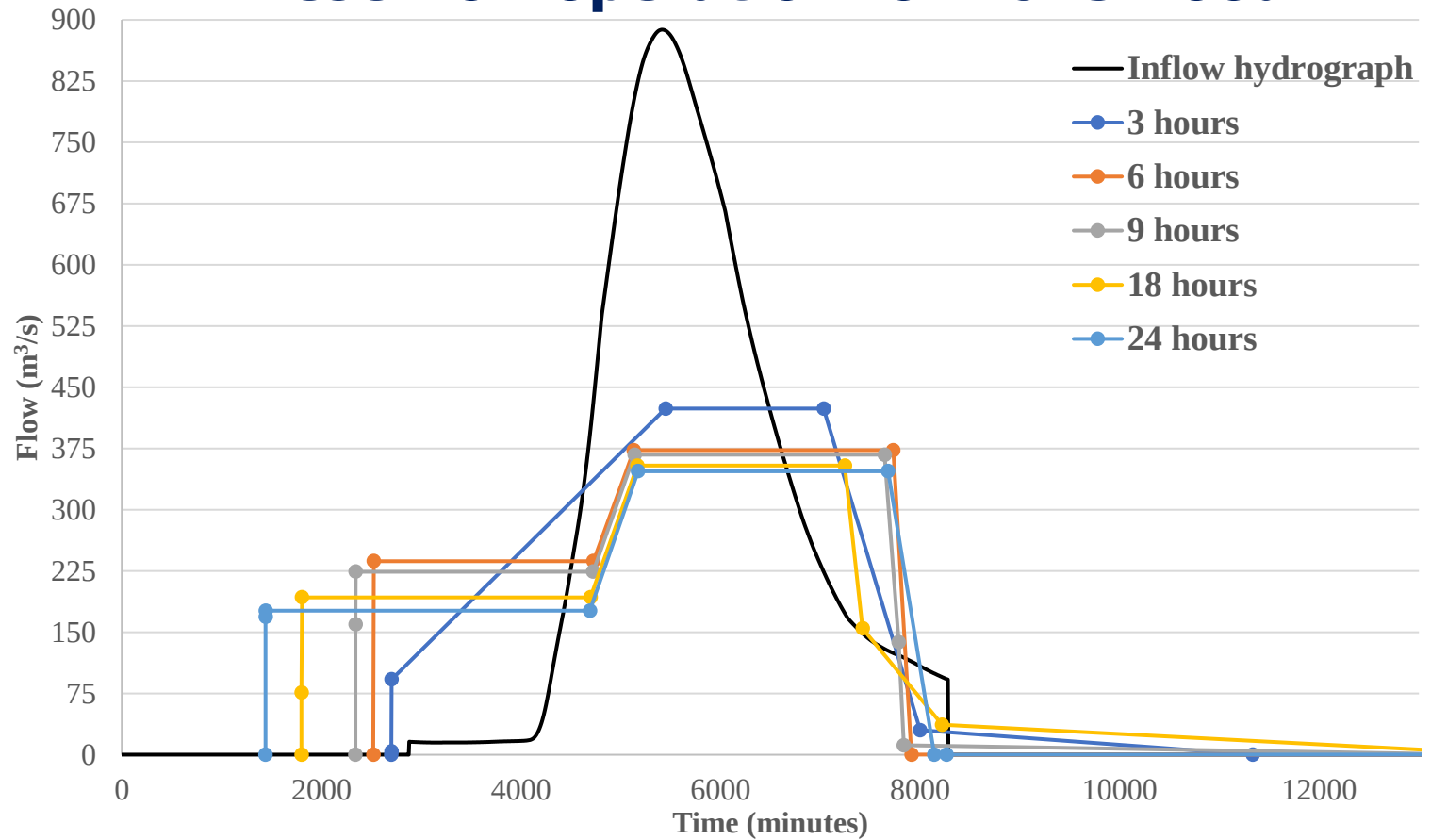
Calibration & Validation of hydrological model



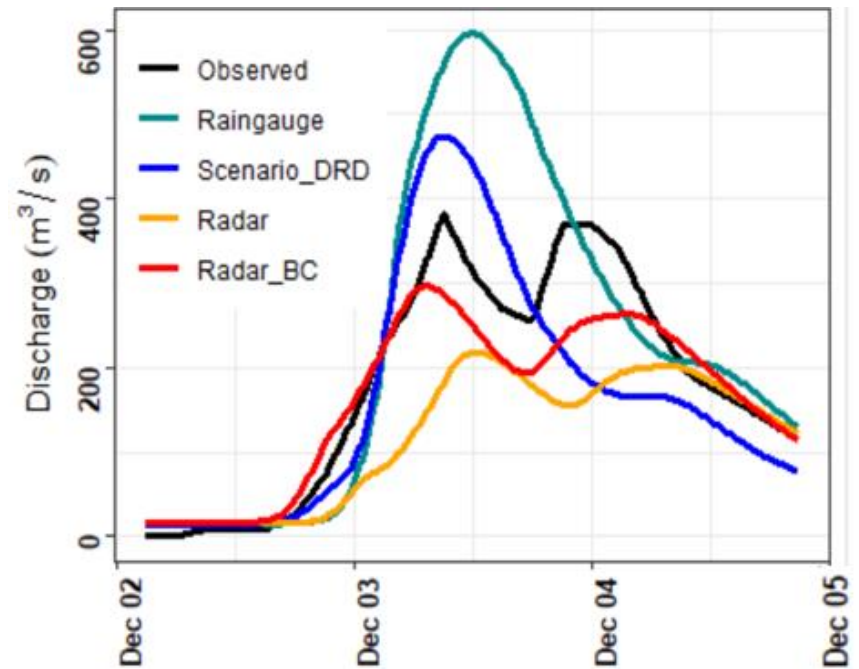
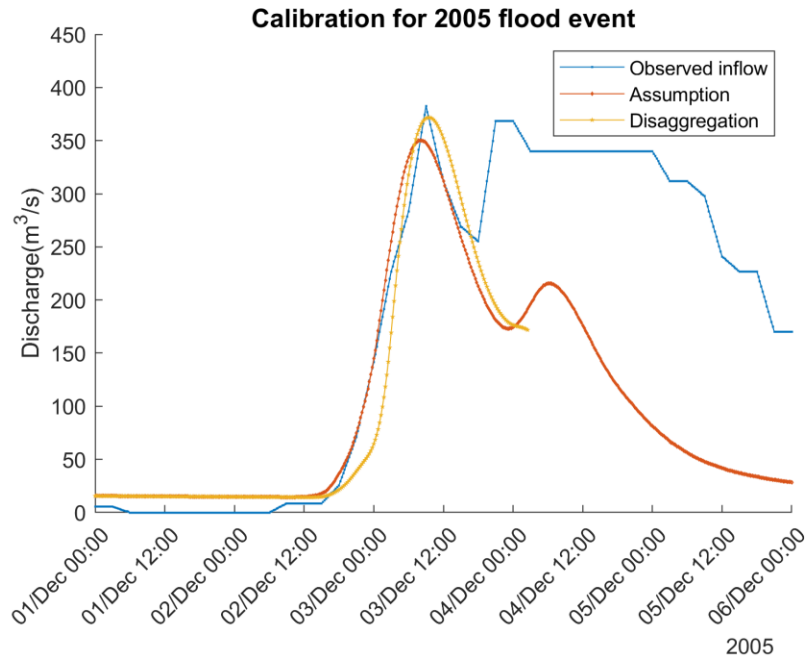
Reservoir operation for 2015 flood



Reservoir operation for 2015 flood



Effect of Spatial variability of Rainfall Events



Using Doppler weather radar for nowcasting would improve the inflow predictions significantly and will be very useful for issuing flood warning



Water Sensitive Urban Design (WSUD)

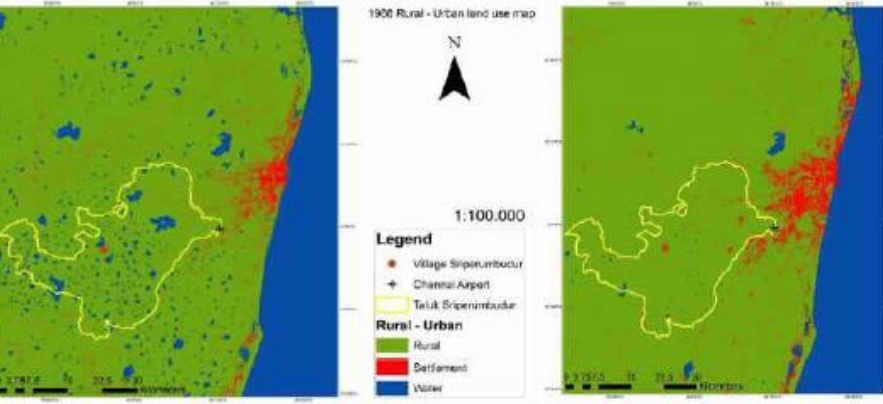
Planning for increasing resilience





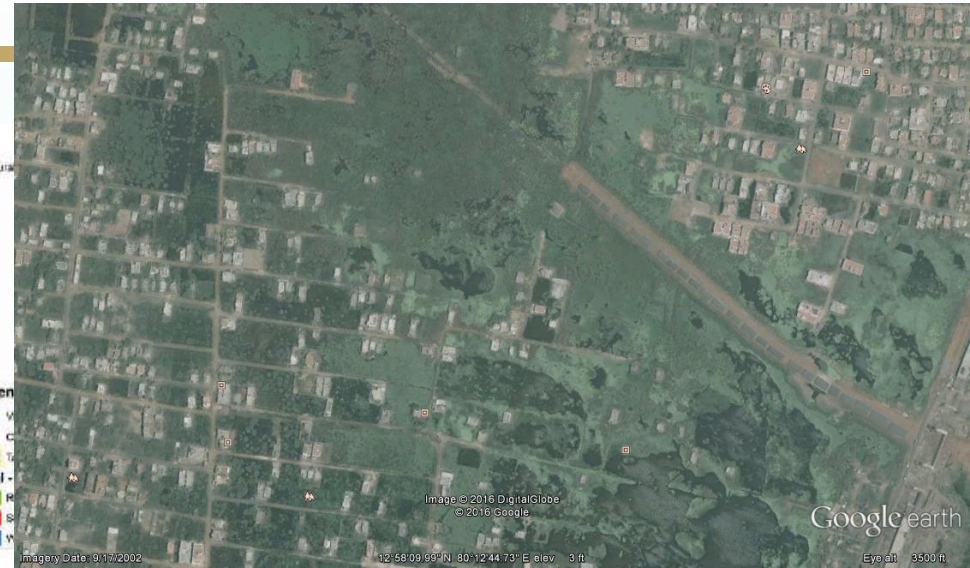
Rapid and Unplanned Urbanization

2002

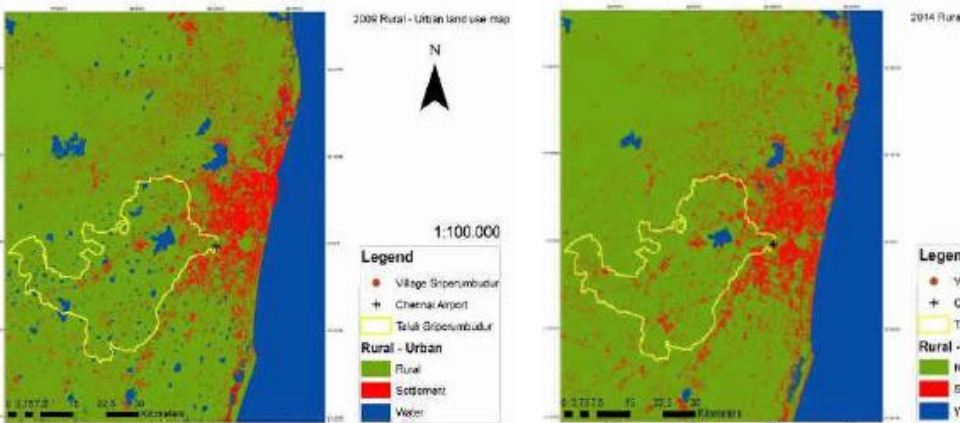


(a) Year 1988

(b) Year 2001



2016



(c) Year 2009

(d) Year 2014

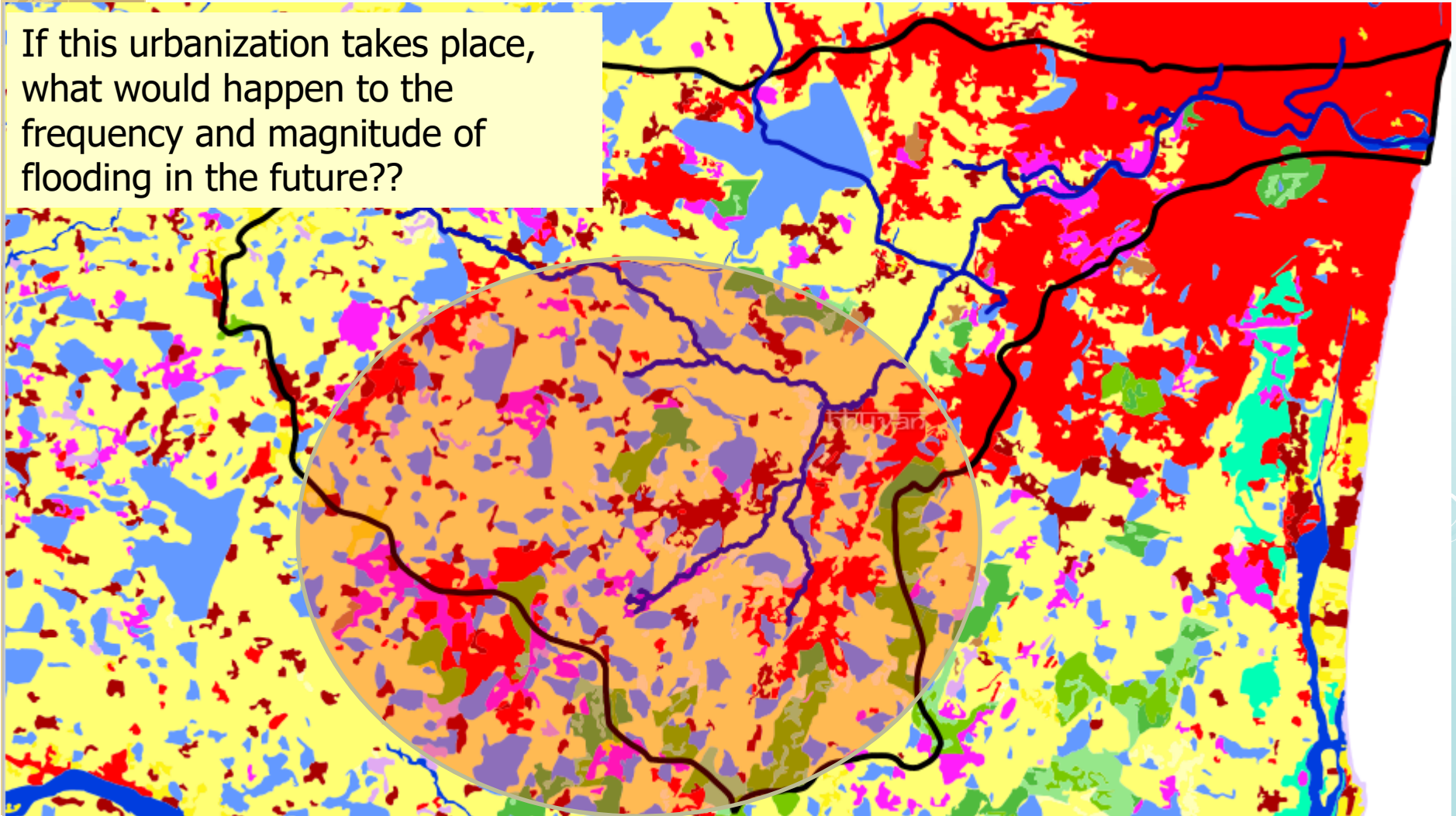


Figure 13: Land use change in Chennai and its suburbs (Murawski, 2016)



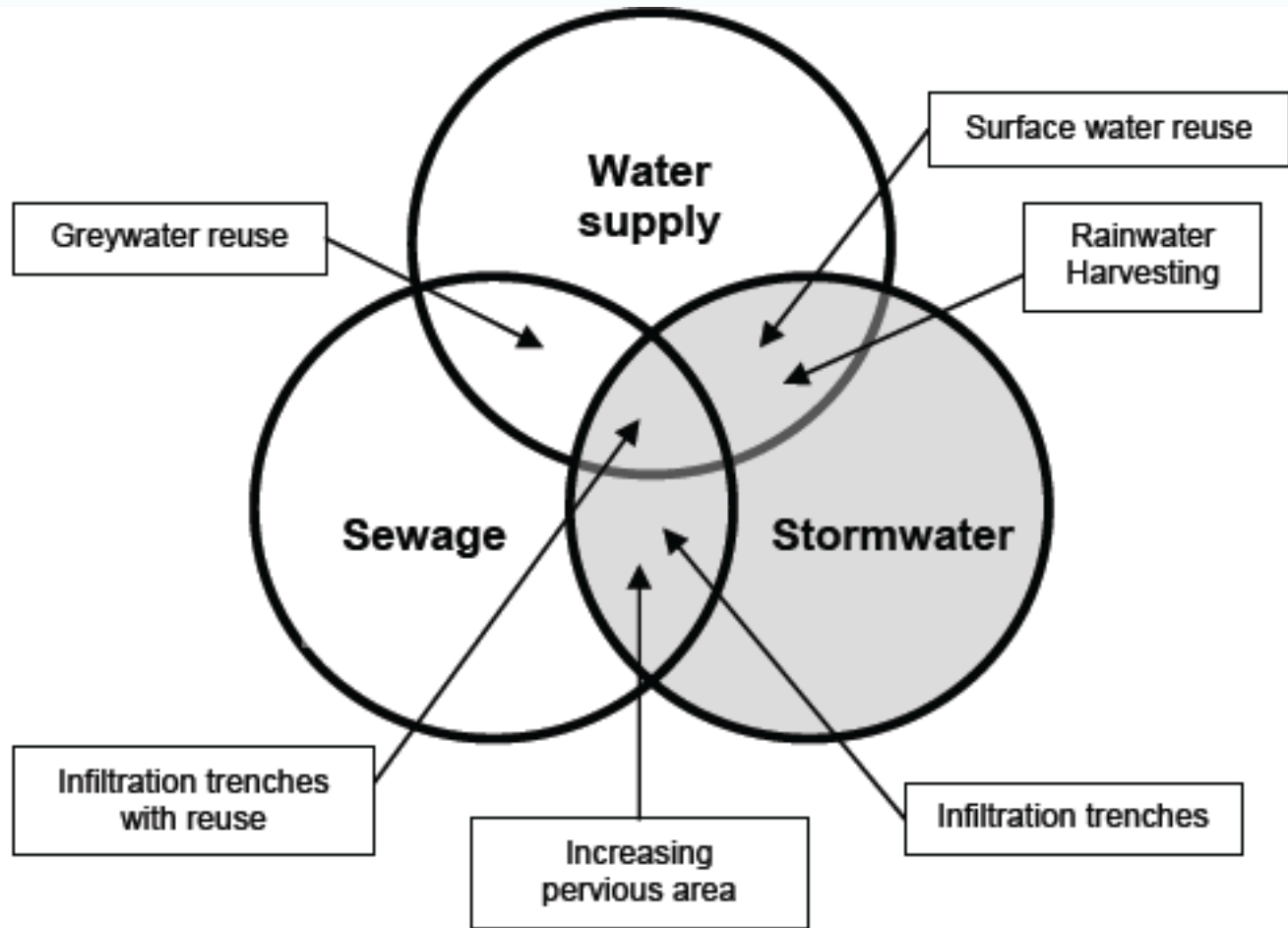
Future scope for urbanization

If this urbanization takes place, what would happen to the frequency and magnitude of flooding in the future??





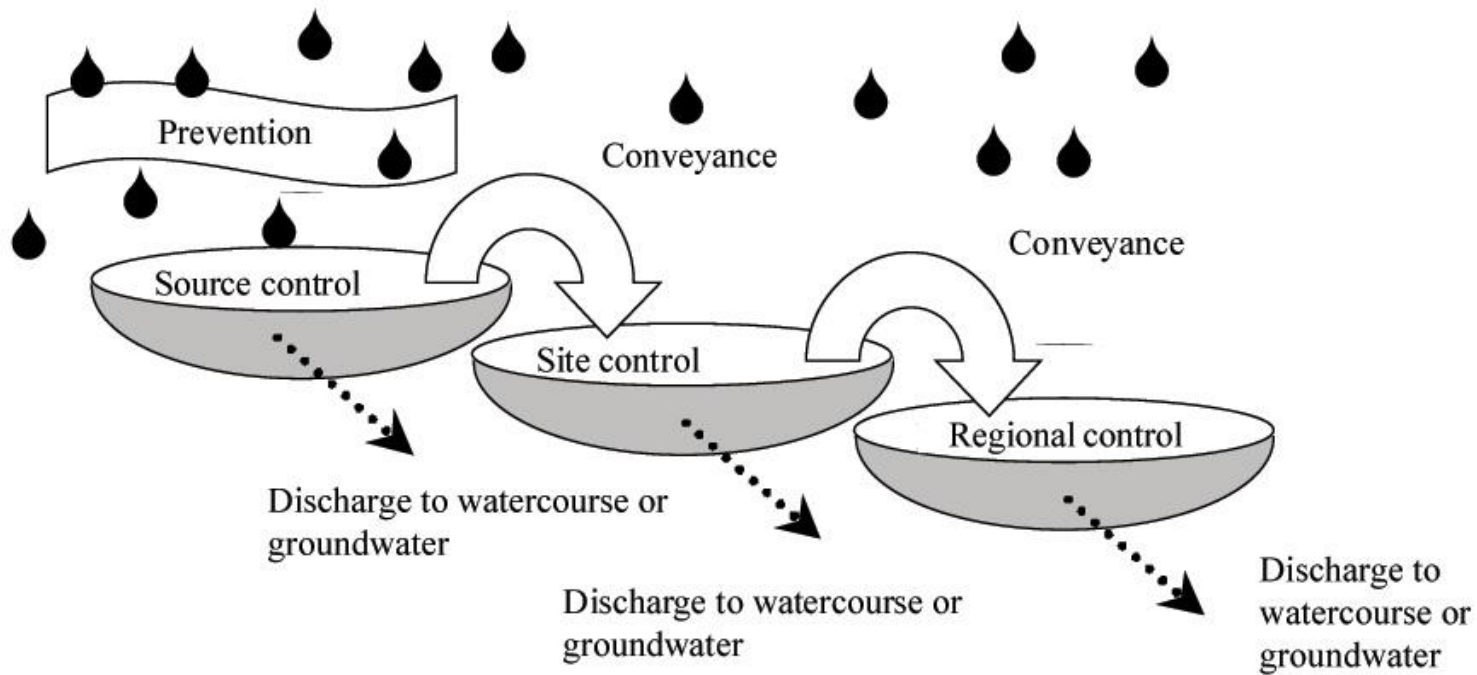
Water Cycle as a whole



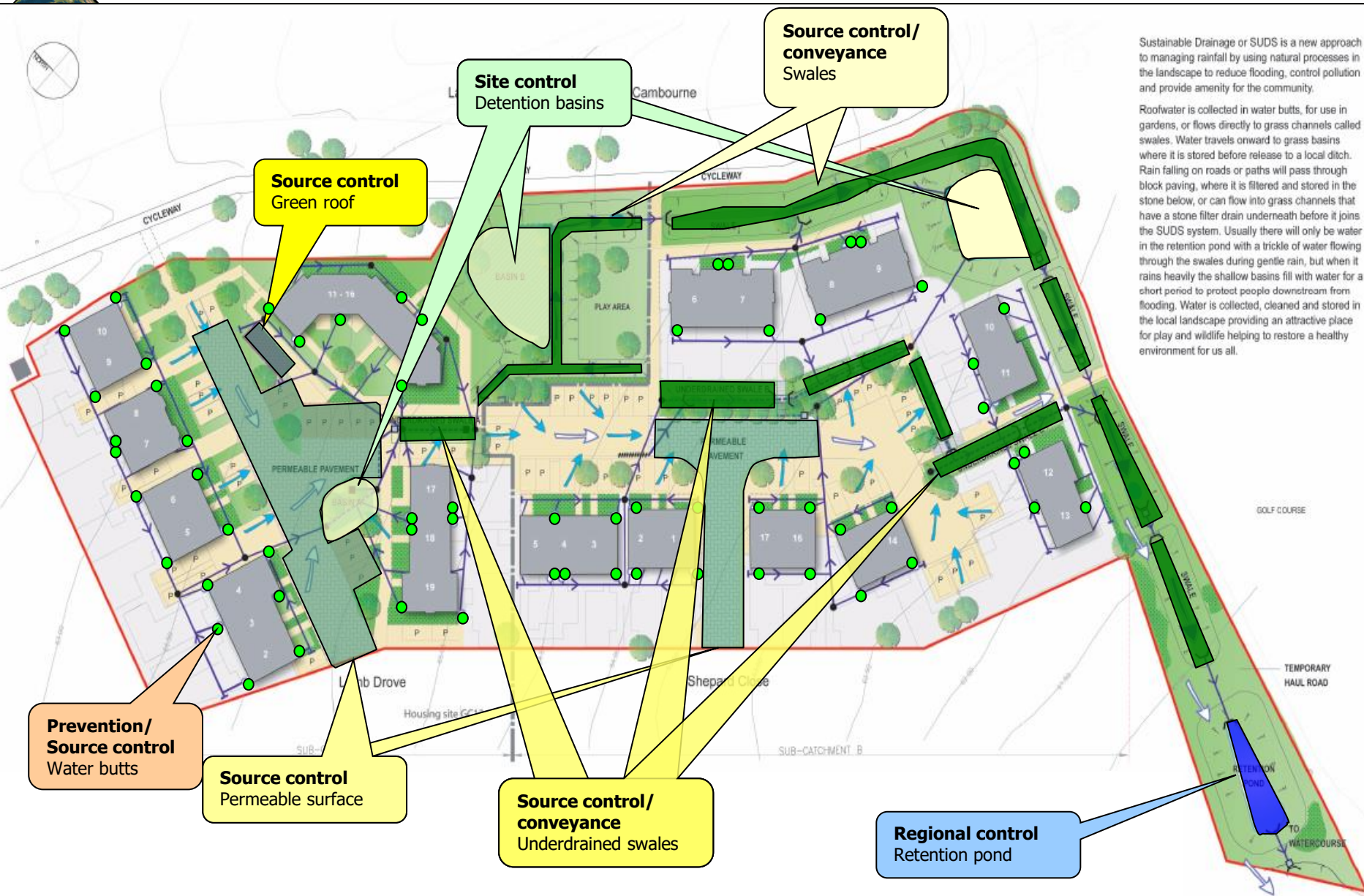


Sustainable Drainage Systems (SUDS)

✚ SuDS management train



Source: CIRIA, 2000
www.susdrain.org



**Prevention/
Source control**
Water butts

Source control
Permeable surface

Source control
Green roof

Site control
Detention basins

**Source control/
conveyance**
Underdrained swales

**Source control/
conveyance**
Swales

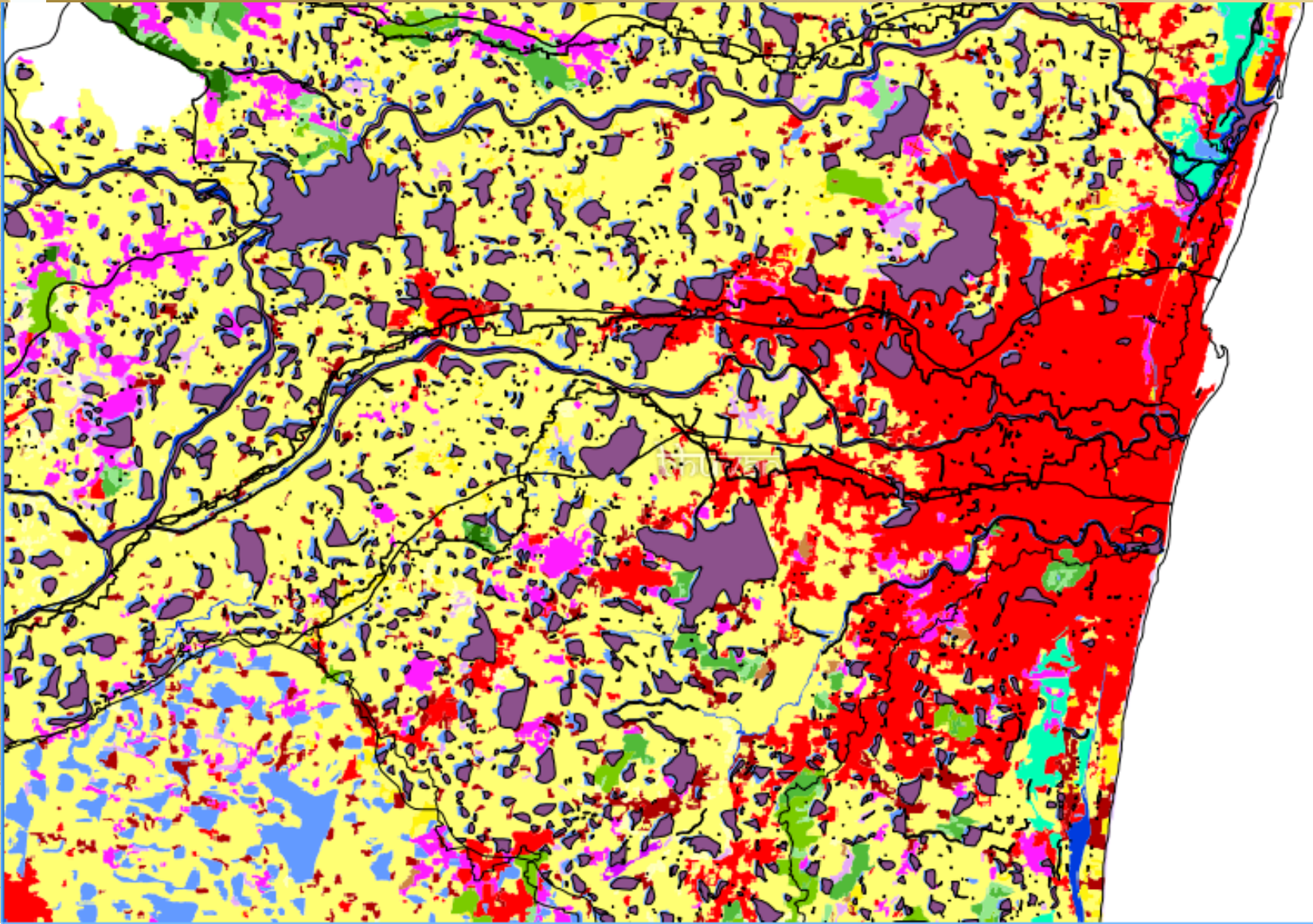
Regional control
Retention pond

Sustainable Drainage or SUDS is a new approach to managing rainfall by using natural processes in the landscape to reduce flooding, control pollution and provide amenity for the community.

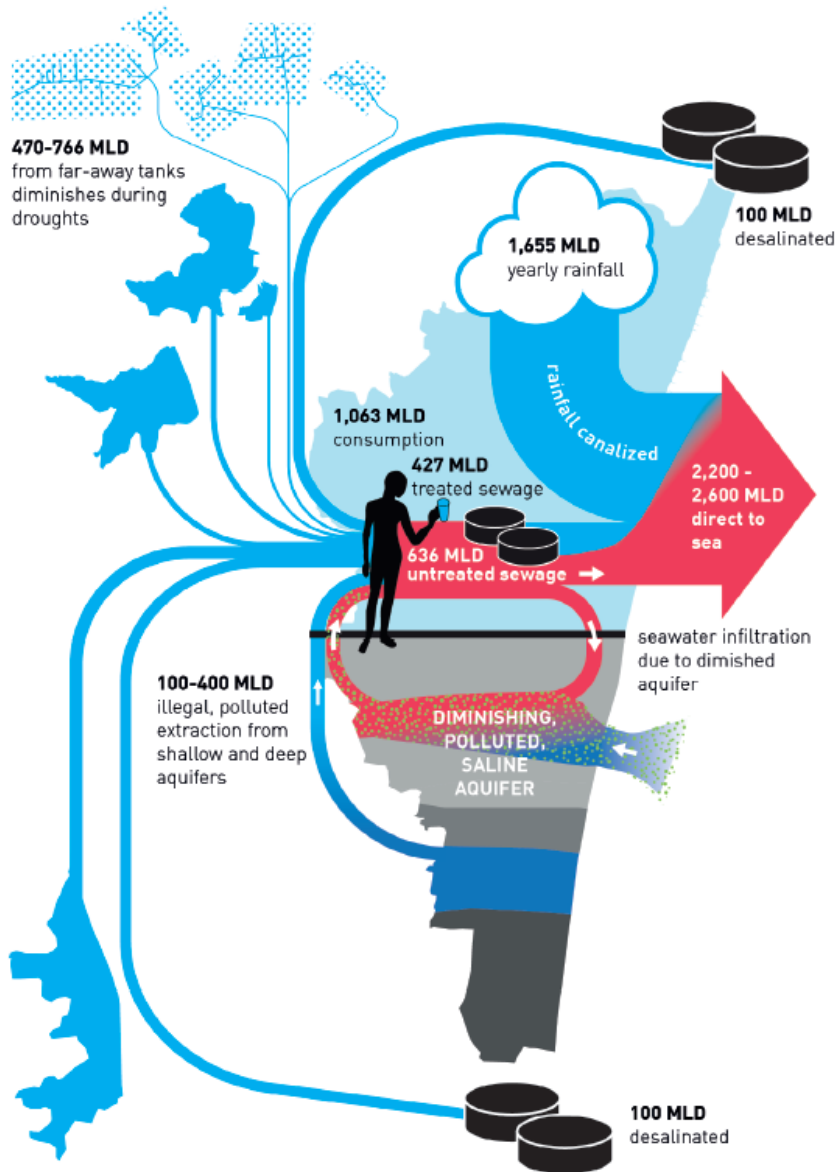
Roofwater is collected in water butts, for use in gardens, or flows directly to grass channels called swales. Water travels onward to grass basins where it is stored before release to a local ditch. Rain falling on roads or paths will pass through block paving, where it is filtered and stored in the stone below, or can flow into grass channels that have a stone filter drain underneath before it joins the SUDS system. Usually there will only be water in the retention pond with a trickle of water flowing through the swales during gentle rain, but when it rains heavily the shallow basins fill with water for a short period to protect people downstream from flooding. Water is collected, cleaned and stored in the local landscape providing an attractive place for play and wildlife helping to restore a healthy environment for us all.



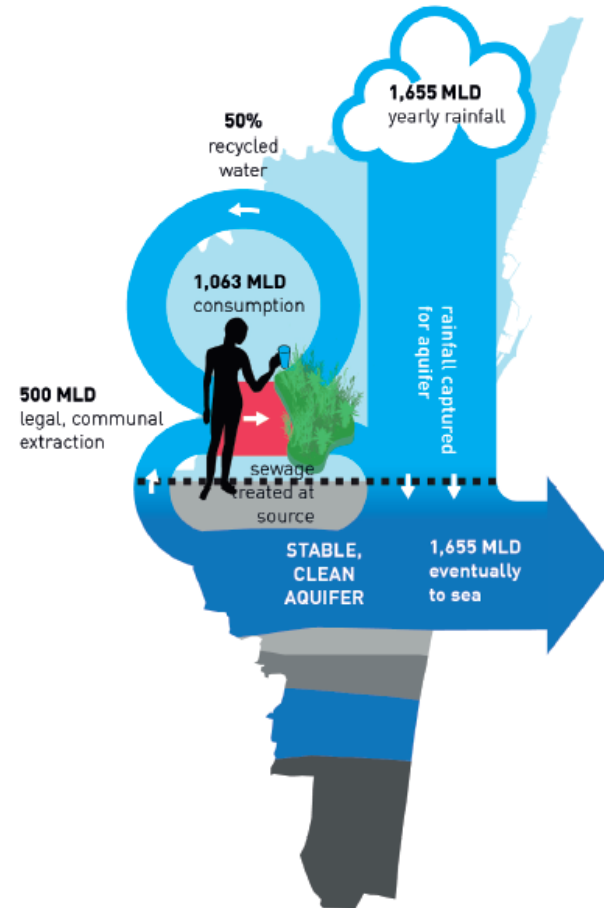
100's of water bodies



CURRENT SYSTEM SCARCITY



PROPOSED SYSTEM CLOSED LOOP

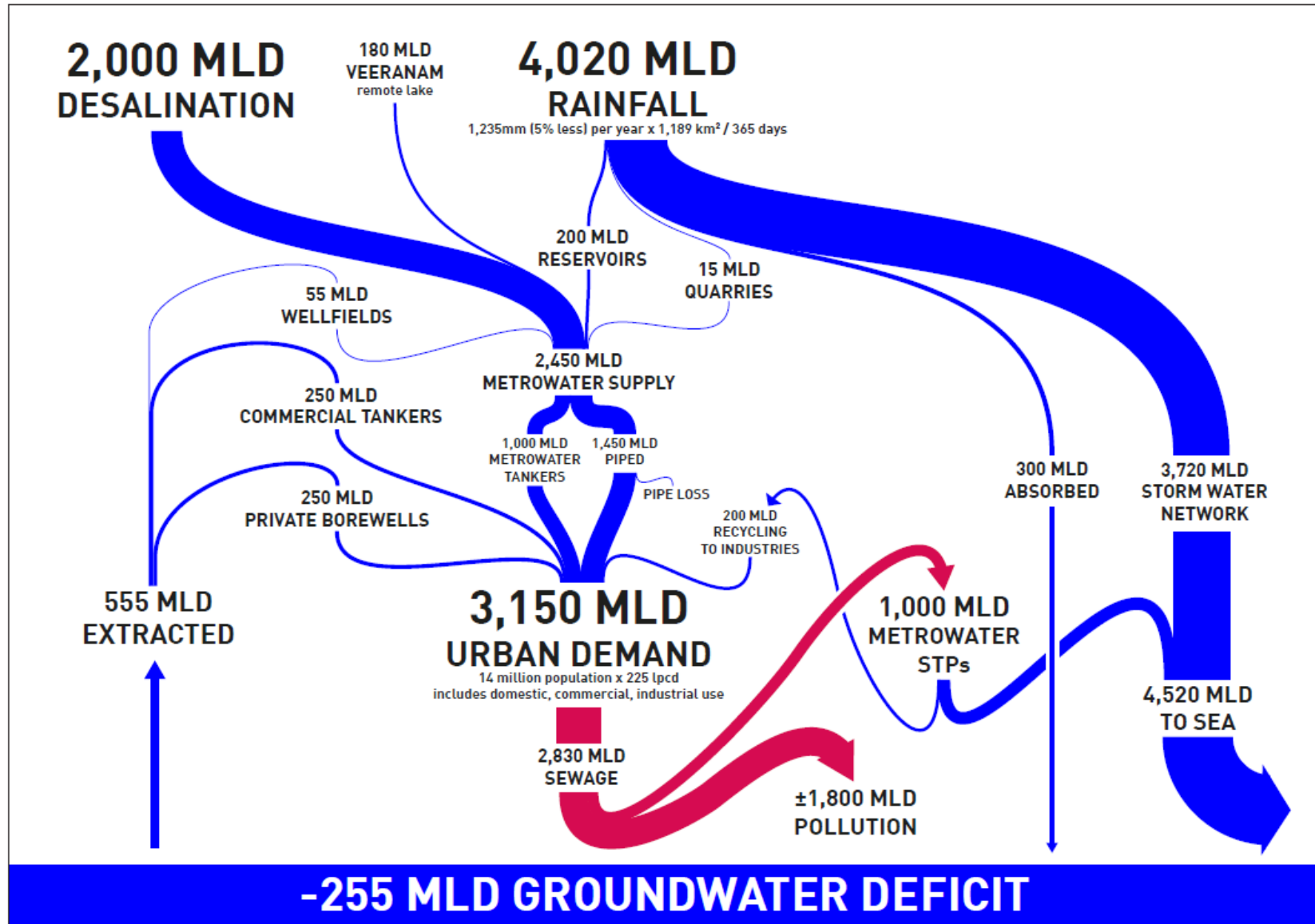


Current recycling: 10%
(only to industries)

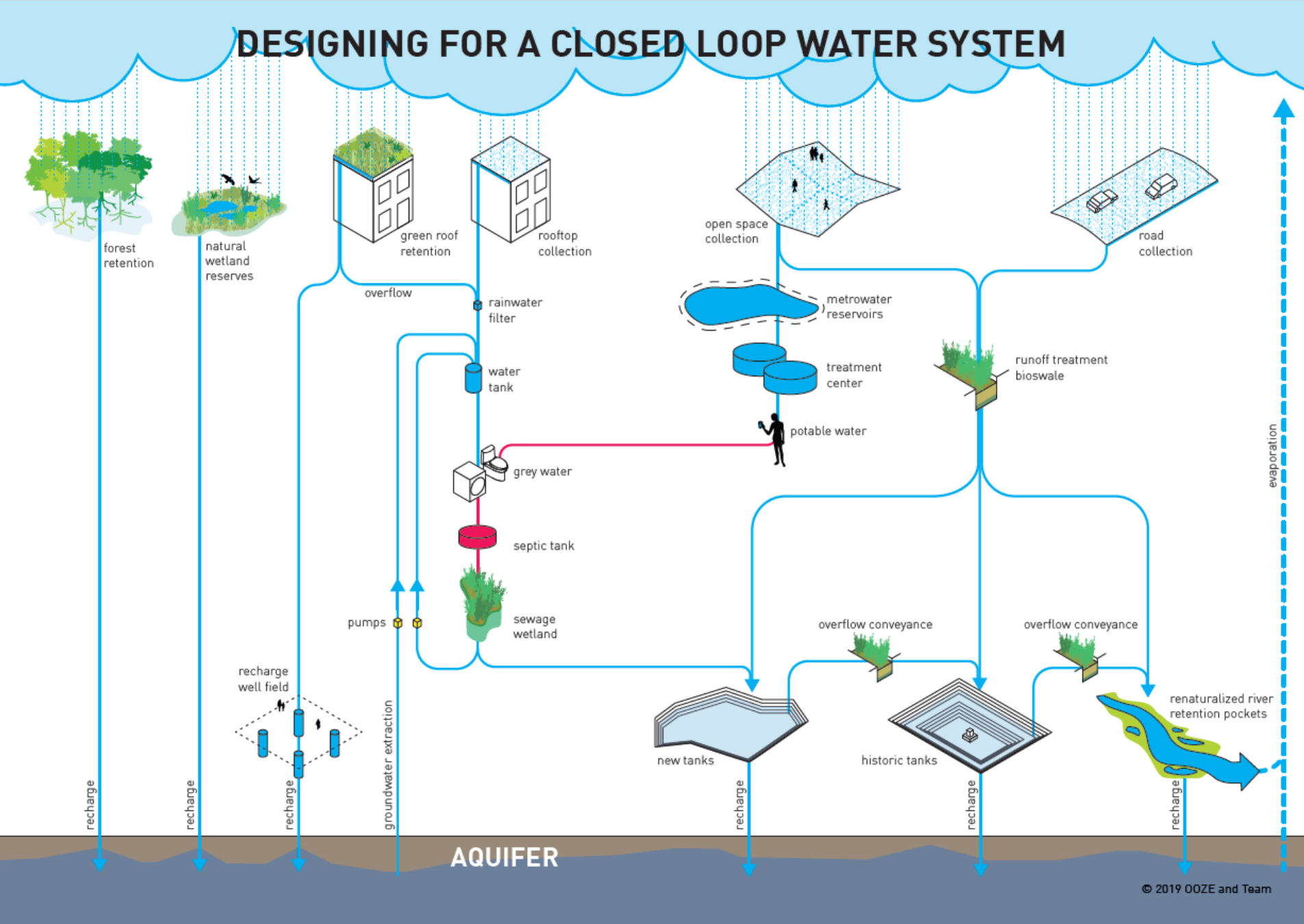
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Source: City of 1000 Tanks

BUSINESS AS USUAL 2030



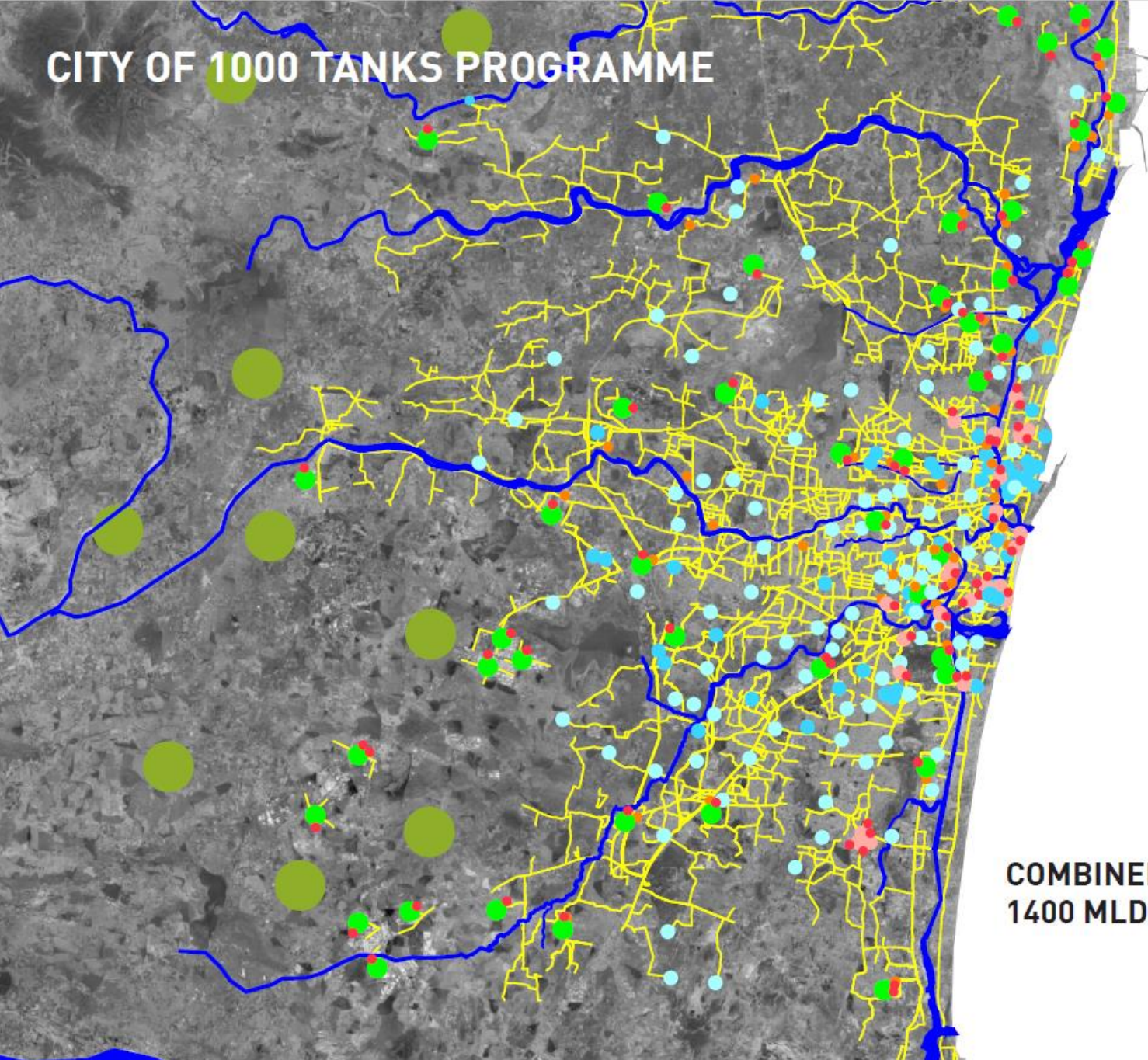
DESIGNING FOR A CLOSED LOOP WATER SYSTEM



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Source: City of 1000 Tanks

CITY OF 1000 TANKS PROGRAMME



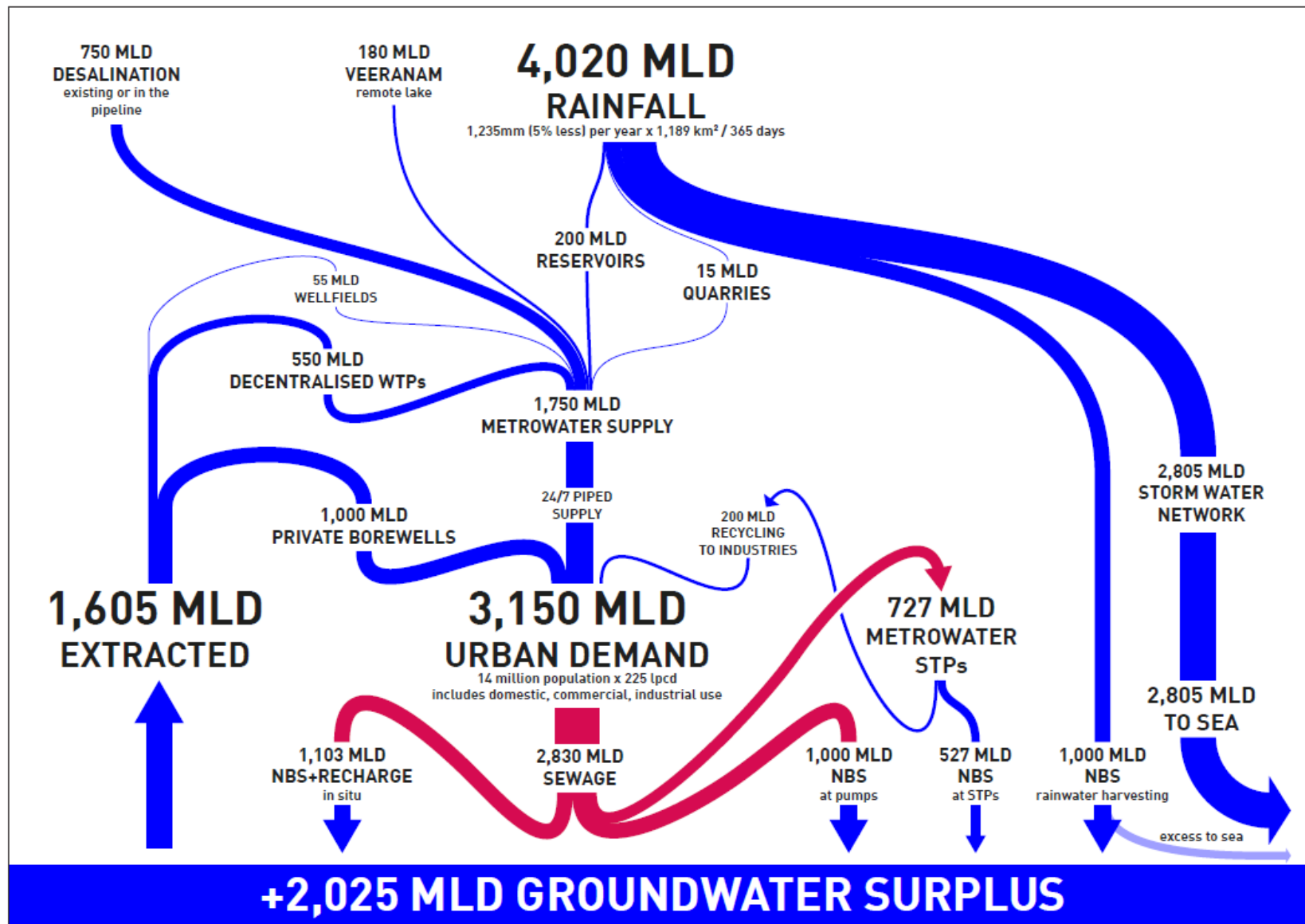
COMBINED PROJECT
1400 MLD recharge

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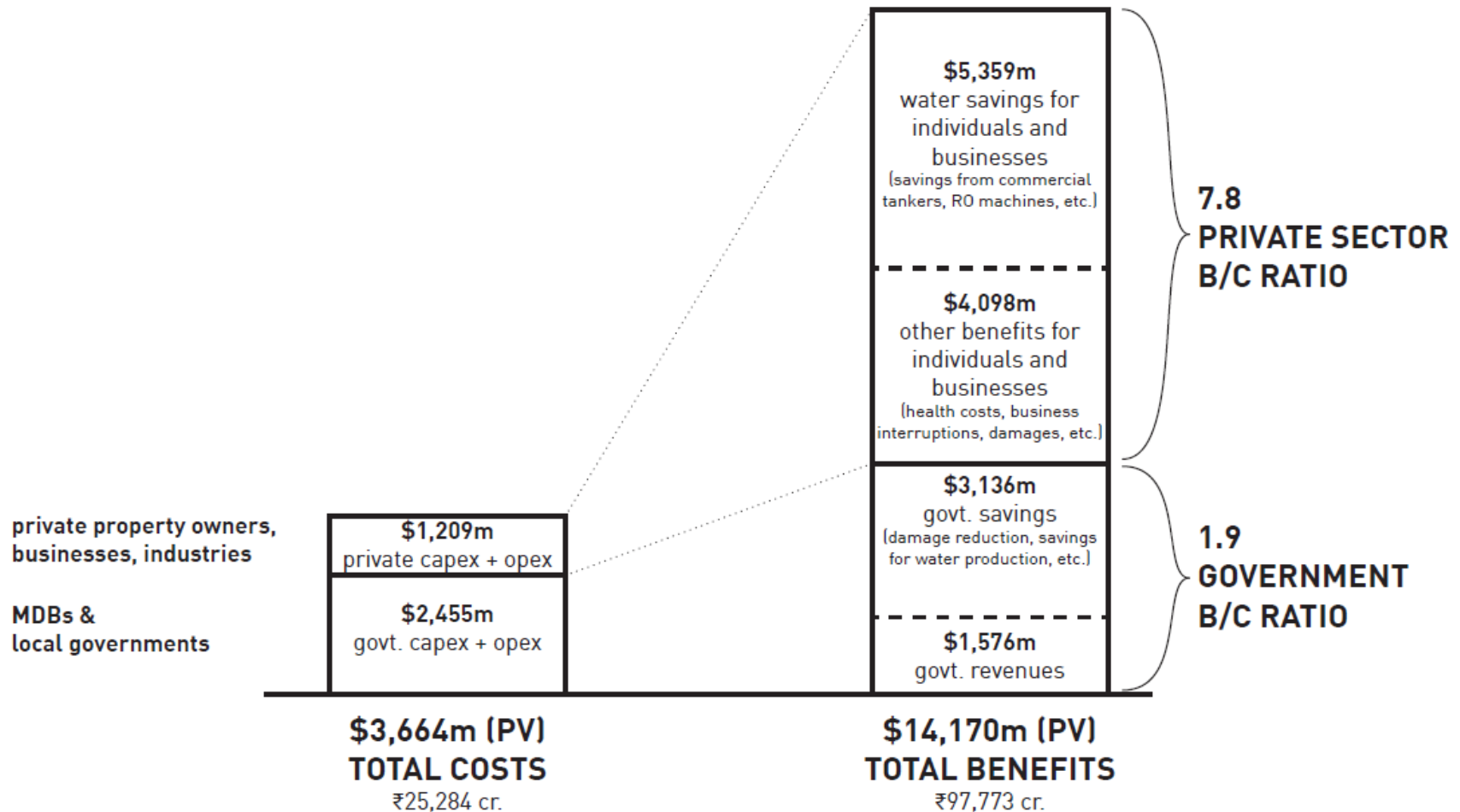
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Source: City of 1000 Tanks

WATER BALANCE MODEL 2030



CITY WIDE PROGRAM B/C RATIO = 3.9



NPV = \$10,560m

₹72,490 cr.

Represented in millions USD Present Value (PV)

Yearly flows calculated with 6% discount rate over a 50 year lifecycle

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Tangible actions

- ⊕ Ensure road resurfacing does not result in aggradation of road level
 - ⊠ Adopt IRC:120-2015
- ⊕ Adequate cross-drainage works across streets/roads and highways
- ⊕ Revive tanks wherever possible
 - ⊠ Where tanks are lost, at least revive the outlet and excess sluice
- ⊕ Demarcate flood lines of rivers, drains and lakes
 - ⊠ Restrict activities within these demarcated water line zones
- ⊕ Proper solid waste management
 - ⊠ Community Education on waste segregation and solid waste management



Tangible actions

- ✚ Mandate Sustainable Urban Drainage Systems (SuDs) for all new developments
 - ✚ Permit system for storm water discharge
 - ✚ Ensure that urbanization does not necessarily result in higher rate of runoff
 - ✚ Macro drainage in place before new development



FLOOD RISK $\rightarrow 0$?

~~CHANCE $\rightarrow 0$?~~

~~Or~~

~~DAMAGE $\rightarrow 0$?~~

✚ But the city will be
resilient to
flooding 😊